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**EXAM REPORT
NON LINEAR DYNAMICS & CONTROL
ELECTRONIC THROTTLE BODY SYSTEM**

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Contents

1	Context	4
1.1	Physical Model	5
1.1.1	Non-linear State Space Representation	6
1.1.2	Linearization (Small Signal Model)	7
2	Open-loop Analysis	8
2.1	Equilibria	8
2.2	Stability	9
2.2.1	Phase Plane Analysis	9
2.2.2	Lyapunov Theory	11
2.2.3	Parameter Variation and Bifurcations	11
3	Synthesis	13
3.1	Control Problem	13
3.2	Linear Control Strategy: State Feedback	14
3.2.1	Controller Design	14
3.2.2	Nominal Results	16
3.3	Nonlinear Control Strategy: Sliding Mode Control	17
3.3.1	Controller Design	17
3.3.2	Nominal Results	19
4	Robustness Analysis	20
4.1	Robustness to Disturbances	20
4.1.1	State Feedback Controller (SFC)	20
4.1.2	Sliding Mode Controller (SMC)	21
4.2	Robustness to Uncertainties	22
4.2.1	State Feedback Controller (SFC)	23
4.2.2	Sliding Mode Controller (SMC)	24
4.3	Combined Scenario (Disturbance + Uncertainty)	25

5	Performance Comparison	26
5.1	Simulation Environment Setup	26
5.2	Tracking Performance	28
5.2.1	Nominal Conditions	28
5.2.2	Robustness (Dynamic Tracking with Disturbances and Un- certainty)	29
5.3	All Scenarios	30
5.3.1	Disturbance with Static Reference	31
5.3.2	Uncertainty with Static Reference	32
5.3.3	Disturbance & Uncertainty with Static Reference	33
5.3.4	Nominal with Dynamic Reference	34
6	Conclusion	35
6.1	Summary of Analysis	35
6.2	Control Strategy Comparison	35
6.3	Control Effort and Chattering	36
6.4	Conclusion on Comparison	36

Introduction

Introduction

The following report was written for the examination of the course ‘Nonlinear Dynamics and Control’. The objective of this report is to study and analyze a nonlinear dynamic system, apply control strategies, and compare the performance obtained with various controllers.

First, in the “Context” chapter, the non-linear system examined is described and the structure of the model is explained. The system under analysis is an **Electronic Throttle Body**, a critical automotive component characterized by non-linearities such as friction and the non-linear stiffness of the return spring. An open-loop analysis is then carried out by applying the non-linear system analysis methods covered in the course (specifically Phase Plane analysis), supported by appropriate simulations.

At which point, the control problem was formulated with its performance specification. The control strategies used were presented: the linear controller implemented is the **State Feedback Controller (SFC)**, while the non-linear controller is the **Sliding Mode Controller (SMC)**. A Simulink model was implemented for both strategies, the results of which were reported.

Once the nominal results were obtained, a robustness analysis was carried out in the presence of external disturbances, parametric uncertainties and also disturbances and uncertainties together for both controllers to evaluate their resilience. Finally, a quantitative comparison was made between the performances of the linear and nonlinear controllers, highlighting the differences in tracking accuracy and control effort.

The Electronic Throttle Body was chosen in order to apply the knowledge developed in this course on a crucial real component in automotive system, for which there is a familiarity obtained during the experience in the Propulsion Hardware-In-loop Team of Stellantis Group.

Chapter 1

Context

The system analysed in this project is an Electronic Throttle Body, a critical component widely used to regulate the air mass flow in internal combustion engines to ensure the stoichiometric ratio. From a mechanical perspective, it can be modelled as a rotational mass-spring-damper system driven by a DC motor. However, strictly non-linearities, particularly regarding the friction (stick-slip phenomena) and the non-linear stiffness of the return spring, introduce complex dynamic behaviours that cannot be treated with linear techniques alone.

One of the fundamental aspects of the study of non-linear systems is to identify the behaviour of the system as physical parameters or initial conditions change. Concepts such as bifurcations, limit cycles, and local/global stability are key in this context and require careful and detailed modelling to ensure precise position tracking of the throttle valve.

1.1 Physical Model

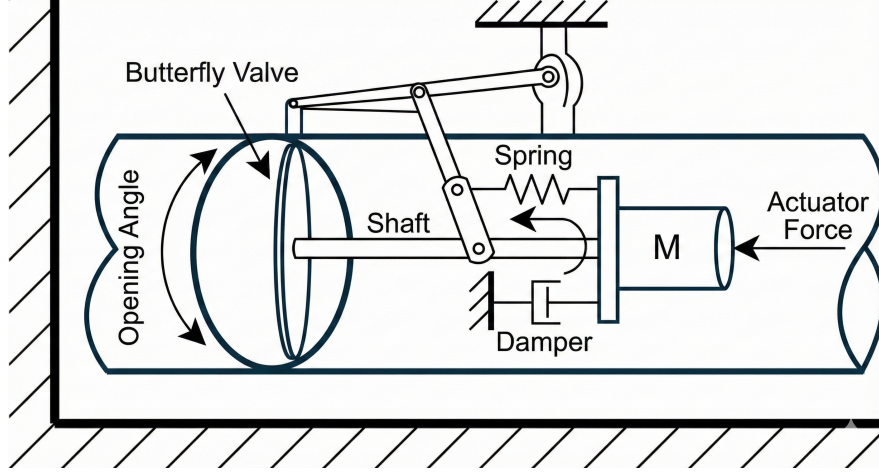


Figure 1.1: Electronic Throttle Body Scheme

Referring to the physical model, the angular position of the valve is denoted by the variable $\theta(t)$, which represents the main state variable, while the angular velocity $\dot{\theta}(t)$ is the second state variable. The mathematical model based on the balance of moments is obtained as follows:

$$J\ddot{\theta}(t) = u(t) - b_1\dot{\theta}(t) - b_2\dot{\theta}(t)|\dot{\theta}(t)| - k_1\theta(t) - k_2\theta^3(t) \quad (1.1)$$

Rearranging the terms, the equation of motion is:

$$J\ddot{\theta} + b_1\dot{\theta} + b_2\dot{\theta}|\dot{\theta}| + k_1\theta + k_2\theta^3 = u \quad (1.2)$$

Where the physical parameters and variables are described below.

Table 1.1: Physical parameters and variables of the Throttle Body system

Symbol	Unit	Description
$\theta(t)$	[rad]	Valve opening angle (Position)
$\dot{\theta}(t)$	[rad/s]	Valve angular velocity
$\ddot{\theta}(t)$	[rad/s ²]	Angular acceleration
J	[kg·m ²]	Total moment of inertia
b_1	[Nms/rad]	Linear viscous friction coefficient
b_2	[Nms ² /rad ²]	Aerodynamic damping coefficient (non-linear)
k_1	[Nm/rad]	Linear elastic constant of the return spring
k_2	[Nm/rad ³]	Cubic non-linearity coefficient of the spring
$u(t)$	[Nm]	Control torque provided by the DC motor

The expression $\dot{\theta}|\dot{\theta}|$ represents the non-linear aerodynamic damping. It is used to preserve the sign of the velocity, ensuring that the resulting resistive force is always directed against the direction of motion:

$$\dot{\theta}|\dot{\theta}| = \dot{\theta}^2 \cdot \text{sign}(\dot{\theta})$$

1.1.1 Non-linear State Space Representation

To perform the analysis and simulation, the system is described by means of a second-order non-linear model. The state variables are chosen as follows:

$$x_1 = \theta \quad x_2 = \dot{\theta}$$

Substituting these into the equation of motion (1.1), the system in the state space is defined as:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\frac{k_1}{J}x_1 - \frac{k_2}{J}x_1^3 - \frac{b_1}{J}x_2 - \frac{b_2}{J}x_2|x_2| + \frac{1}{J}u \end{cases} \quad (1.3)$$

1.1.2 Linearization (Small Signal Model)

For the design of the linear controller (State Feedback), the system is linearized around an equilibrium point, defined as the *Limp Home* position θ_{eq} . The state variables are defined as deviations from this equilibrium:

$$x_1 = \delta\theta = \theta - \theta_{eq} \quad (1.4)$$

$$x_2 = \delta\dot{\theta} = \dot{\theta} \quad (1.5)$$

The equivalent stiffness of the spring at the linearization point, denoted as K_{eq} , is calculated as:

$$K_{eq} = \left. \frac{\partial \tau_{spring}}{\partial \theta} \right|_{\theta_{eq}} = k_1 + 3k_2\theta_{eq}^2 \quad (1.6)$$

Neglecting higher-order terms, the linearized equation of motion becomes:

$$J\delta\ddot{\theta} + b_1\delta\dot{\theta} + K_{eq}\delta\theta = \delta u \quad (1.7)$$

Consequently, the linearized system can be expressed in the standard state-space matrix form:

$$\begin{cases} \dot{\mathbf{x}}(t) = A\mathbf{x}(t) + Bu(t) \\ y(t) = C\mathbf{x}(t) + Du(t) \end{cases} \quad (1.8)$$

By substituting the physical parameters, the system matrices are derived as:

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{K_{eq}}{J} & -\frac{b_1}{J} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{1}{J} \end{bmatrix} \quad (1.9)$$

$$C = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad D = 0 \quad (1.10)$$

Chapter 2

Open-loop Analysis

In this chapter, an analysis of the dynamics of the system under consideration is carried out. First, the equilibrium point is identified, followed by a stability analysis using the Phase Plane method (covered in the course). Finally, a study of the system's behavior under parameter variations is presented to investigate potential bifurcations or topological changes.

2.1 Equilibria

Finding the equilibrium points is the first step. Starting with the system obtained in the previous chapter and expressed by the state-space equation (1.3), considering the system with no input ($u = 0$) and assuming the stationary condition $\dot{x}_1 = 0$ and $\dot{x}_2 = 0$, the equilibrium condition is obtained from the second equation:

$$0 = -\frac{k_1}{J}x_1 - \frac{k_2}{J}x_1^3 \Rightarrow -\frac{x_1}{J}(k_1 + k_2x_1^2) = 0 \quad (2.1)$$

From which it follows that the roots are:

- $x_1 = 0$ (the origin);
- $x_1 = \pm\sqrt{-k_1/k_2}$.

Since the stiffness coefficients are physically defined as positive ($k_1 > 0, k_2 > 0$), the term under the square root is negative. Consequently, there are no real non-zero equilibrium points. The system possesses a **unique equilibrium point**, which is the origin ($x_1 = 0, x_2 = 0$).

2.2 Stability

At this point, the type of stability that characterises the equilibrium point is studied using two methods covered during the course.

2.2.1 Phase Plane Analysis

The first method used is the Phase Plane analysis. It is a very useful tool, which makes it possible to obtain a complete classification of all equilibria in the planar system by studying the Jacobian matrix, in particular its eigenvalues.

Considering a generic planar system and given its dynamic matrix A , the eigenvalues are solutions of the characteristic equation:

$$\lambda^2 - \tau\lambda + \Delta = 0 \quad (2.2)$$

where $\tau = \text{trace}(A)$ and $\Delta = \det(A)$. The solutions are:

$$\lambda_{1,2} = \frac{\tau \pm \sqrt{\tau^2 - 4\Delta}}{2} \quad (2.3)$$

To apply this tool to the Throttle Body system, the Jacobian matrix is calculated from the non-linear state equations:

$$J_{jac}(x_1, x_2) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k_1}{J} - \frac{3k_2}{J}x_1^2 & -\frac{b_1}{J} - \frac{2b_2}{J}|x_2| \end{bmatrix} \quad (2.4)$$

Evaluating the Jacobian at the unique equilibrium point (the origin), the matrix becomes:

$$J_{jac}(0, 0) = \begin{bmatrix} 0 & 1 \\ -\frac{k_1}{J} & -\frac{b_1}{J} \end{bmatrix} \quad (2.5)$$

The eigenvalues are the solutions of the characteristic equation:

$$\lambda^2 + \frac{b_1}{J}\lambda + \frac{k_1}{J} = 0 \quad (2.6)$$

Here, $\tau = -b_1/J$ and $\Delta = k_1/J$. Given the positivity of the physical parameters ($b_1, k_1, J > 0$):

- $\tau < 0$: The real part of the eigenvalues is negative, indicating a **stable object**.
- $\Delta > 0$: The determinant is positive.

To distinguish between a *stable node* and a *stable spiral*, the discriminant $D = \tau^2 - 4\Delta$ must be evaluated:

$$D = \left(\frac{b_1}{J}\right)^2 - 4\frac{k_1}{J}$$

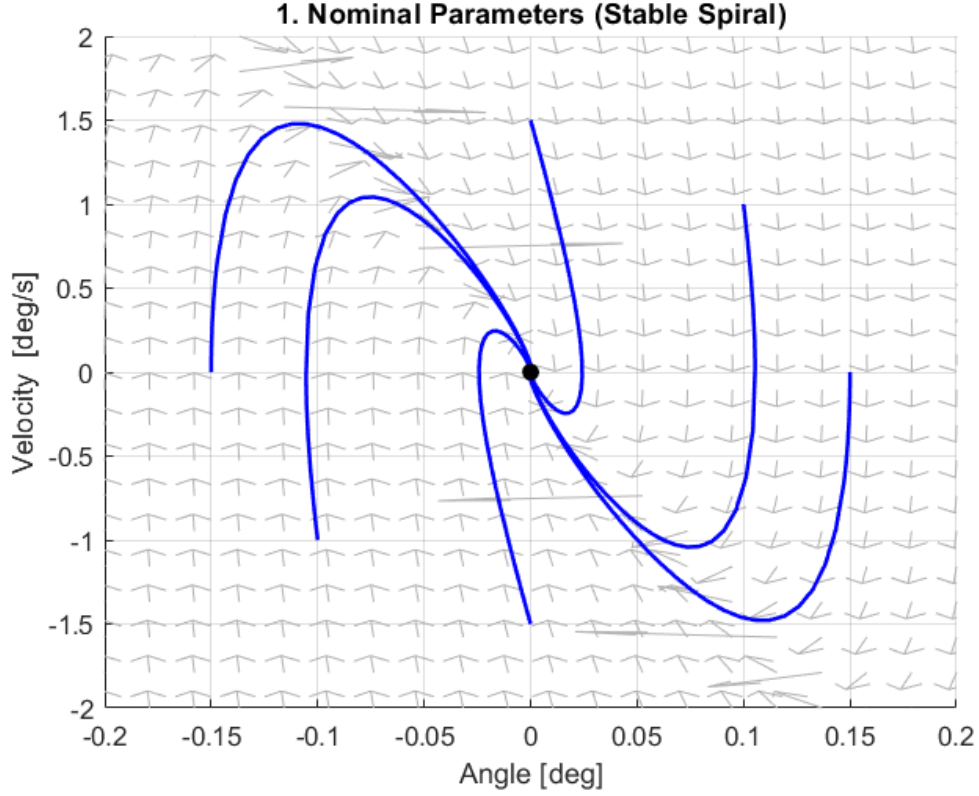


Figure 2.1: Phase Portrait analysis: Nominal Parameters

The simulations performed confirm the theoretical analysis. The vector field of the system, highlights the global attraction towards the origin and the trajectories converge to the origin with an oscillatory behavior. This confirms that for the nominal values, $D < 0$, resulting in complex conjugate eigenvalues. The origin is a **stable spiral (focus)**.

2.2.2 Lyapunov Theory

To rigorously confirm the stability of the origin including the non-linear terms, Lyapunov theory is applied. A candidate Lyapunov function representing the total energy of the system (rotational kinetic energy + potential elastic energy) is chosen:

$$V(x_1, x_2) = \frac{1}{2}k_1x_1^2 + \frac{1}{4}k_2x_1^4 + \frac{1}{2}Jx_2^2 \quad (2.7)$$

This function is positive definite ($V(0) = 0$ and $V(x) > 0$ for $x \neq 0$). Its time derivative along the system trajectories is:

$$\dot{V} = \frac{\partial V}{\partial x_1}\dot{x}_1 + \frac{\partial V}{\partial x_2}\dot{x}_2 = (k_1x_1 + k_2x_1^3)x_2 + Jx_2\dot{x}_2 \quad (2.8)$$

Substituting the equation of motion (1.1) with $u = 0$:

$$\dot{V} = (k_1x_1 + k_2x_1^3)x_2 + x_2(-k_1x_1 - k_2x_1^3 - b_1x_2 - b_2x_2|x_2|) \quad (2.9)$$

$$\dot{V} = -b_1x_2^2 - b_2x_2^2|x_2| \quad (2.10)$$

Since b_1 and b_2 are positive, $\dot{V}$ is strictly negative for all $x_2 \neq 0$ and equals zero only when the velocity is zero. Invoking La Salle's Invariance Principle, it can be concluded that the origin is **asymptotically stable**.

2.2.3 Parameter Variation and Bifurcations

Finally, the behavior of the system under parameter variation was analyzed to identify potential bifurcations or topological changes in the phase portrait. The eigenvalues are given by:

$$\lambda_{1,2} = \frac{-\frac{b_1}{J} \pm \sqrt{(\frac{b_1}{J})^2 - 4\frac{k_1}{J}}}{2} \quad (2.11)$$

Since $b_1, J, k_1 > 0$, the real part of the eigenvalues ($\text{Re}(\lambda) = -b_1/2J$) remains strictly negative. Therefore, the system does not undergo stability bifurcations (e.g., Hopf bifurcation) where eigenvalues cross the imaginary axis. However, topological changes occur:

- **Increasing b_1 :** By increasing the linear damping b_1 , the discriminant may become positive ($\tau^2 > 4\Delta$). The eigenvalues become real and distinct negative. The system transitions from a stable spiral to a **stable node**, eliminating oscillations (overdamped behavior).

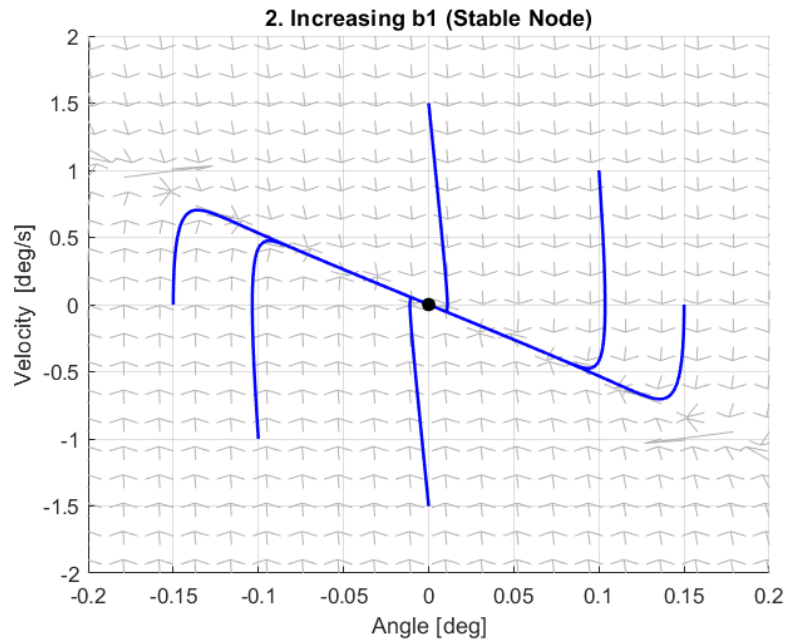


Figure 2.2: Phase Portrait analysis: Increasing $b_1 = 0.03$

- **Decreasing k_1 :** Decreasing the stiffness reduces the natural frequency. The equilibrium remains a stable node, but the trajectories become wider and slower.

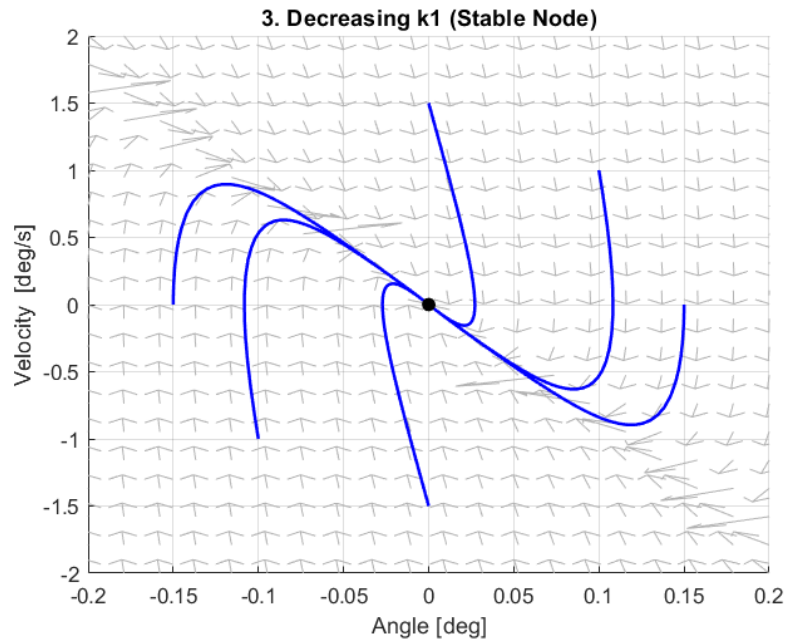


Figure 2.3: Phase Portrait analysis: decreasing $k_1 = 0.075$

Chapter 3

Synthesis

In this chapter, the control problem is formulated with its performance specifications, and the control strategies used are presented. A synthesis of a linear control strategy is first carried out, designed on the linearized model (about the equilibrium point) but applied to the full non-linear system; in this case, a **State Feedback Controller (SFC)** is presented. Subsequently, a synthesis of a non-linear control strategy is detailed; specifically, a **Sliding Mode Controller (SMC)** is developed. All control strategies are supported by simulations showing the evolution of states, tracking error, and control input, carried out using MATLAB/Simulink. The block diagrams of the implemented controllers are also reported.

3.1 Control Problem

The control problem is formulated as a tracking problem where, given a desired valve opening angle (reference θ_{ref}), the controller must ensure the throttle plate reaches that position, minimizing the error between the actual and desired angle. The following performance specifications are considered:

- Settling time less than or equal to 0.15 seconds;
- Percent overshoot less than 13%.

Additionally, robustness against parameter variations (such as changes in spring stiffness or friction) and external disturbances is a key requirement.

3.2 Linear Control Strategy: State Feedback

The following is a synthesis of the linear control strategy using a State Feedback Controller (SFC). This controller is designed based on the linearized model derived in the previous chapter but is applied to the non-linear plant.

3.2.1 Controller Design

First, the controllability of the linearized system around the equilibrium point (Limp Home position) is verified. The state-space matrices are:

$$A = \begin{bmatrix} 0 & 1 \\ -K_{eq}/J & -b_1/J \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 1/J \end{bmatrix} \quad (3.1)$$

The controllability matrix $\mathcal{C} = [B \quad AB]$ is full rank, confirming the system is controllable. The control law is defined as:

$$u_{SFC} = -K\mathbf{x} + K_r r \quad (3.2)$$

Where $K = [k_1 \quad k_2]$ is the feedback gain matrix calculated (e.g., via Pole Placement) to meet the settling time and overshoot requirements, and K_r is a feedforward scalar gain used to ensure zero steady-state error for step inputs.

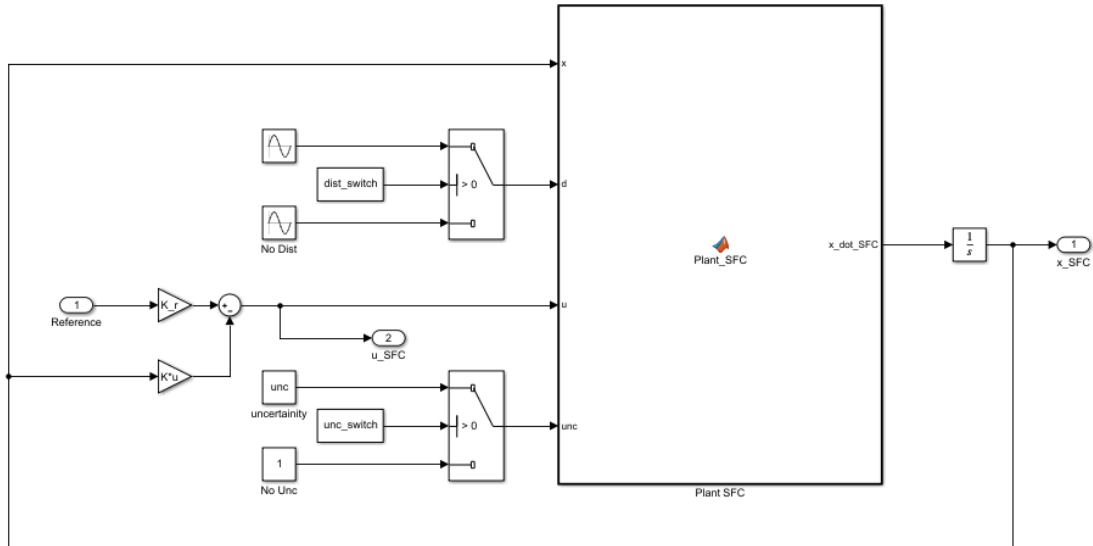


Figure 3.1: Simulink Block Diagram: State Feedback Controller

Here is also reported the MatLab function showed in the SFC diagram to implement the plant.

```

1 function x_dot_SFC = Plant_SFC(x,d,u,unc)
2 % Parameters definition with Uncertainty injection
3 J = 2.5e-4 * unc;
4 k1 = 0.15 * unc;
5 k2 = 0.5 * unc;
6 b1 = 0.01 * unc;
7 b2 = 0.001 * unc;
8
9 % System States
10 x1 = x(1); % Throttle Angle (theta) [rad]
11 x2 = x(2); % Angular Velocity (theta_dot) [rad/s]
12
13 % State Equation 1: Kinematics
14 x1_dot = x2;
15
16 % State Equation 2: Rotational Dynamics
17 % Balance of Moments: J * alpha = Torque_Motor +
    Disturbance - Stiffness - Damping
18 % x2_dot = ( u + d - (k1*x1 + k2*x1^3) - (b1*x2 + b2*x2*abs
    (x2)) ) / J
19
20 x2_dot = - (k1/J)*x1 - (k2/J)*x1^3 - (b1/J)*x2 - (b2/J)*x2*
    abs(x2) + (u + d)/J;
21
22 x_dot_SFC = [x1_dot; x2_dot];
23 end

```

Listing 3.1: MATLAB function describing the Plant dynamics used in the SFC block

The function shows clearly the possibility to inject disturbance and uncertainty without changing anything. In fact, through the external mask of controllers, user can inject both scenarios defining the parameters.

In case of no disturbances $d = 0$ while if there is no uncertainty $unc = 1$.

3.2.2 Nominal Results

The simulation results for the nominal case are shown below.

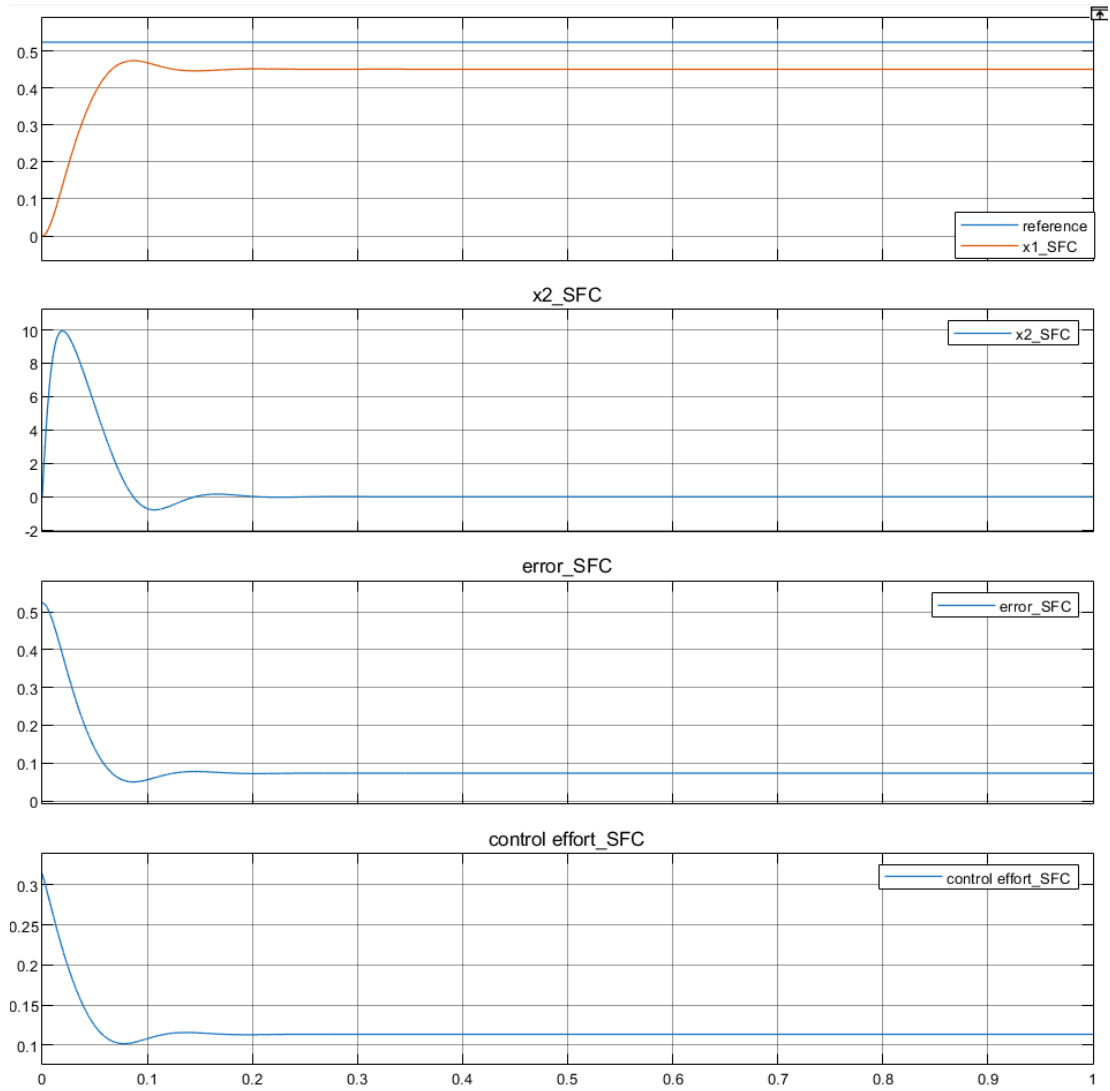


Figure 3.2: SFC Performance (Nominal): Top to bottom - Position tracking (x_1), Velocity (x_2), Tracking Error, Control Effort (u).

The SFC achieves the desired position, but as expected for a linear controller on a non-linear plant, small steady-state errors or deviations may occur when operating away from the linearization point, or due to the uncompensated non-linear friction.

3.3 Nonlinear Control Strategy: Sliding Mode Control

In this section, a Sliding Mode Controller (SMC) is synthesized to explicitly handle the system non-linearities and provide robustness. SMC is designed to force the system trajectories to reach and stay on a specific manifold in the state space, called the *sliding surface*.

3.3.1 Controller Design

The system dynamics can be written in the affine form:

$$\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})u \quad (3.3)$$

where $f(\mathbf{x})$ contains the non-linear spring and damping terms, and $g(\mathbf{x}) = [0, 1/J]^T$.

A linear sliding surface σ is defined based on the tracking error $e = \theta - \theta_{ref}$:

$$\sigma = p_1(x_1 - x_d) + p_2x_2 \quad (3.4)$$

where $c > 0$ determines the convergence rate of the error dynamics once on the surface.

The control law u_{SMC} consists of an equivalent control term u_{eq} (to cancel known dynamics) and a switching term u_{sw} (to handle uncertainty and enforce sliding):

$$u_{SMC} = u_{eq} + u_{sw} \quad (3.5)$$

$$u_{sw} = -K_{sw}\text{sign}(\sigma) \quad (3.6)$$

The gain K_{sw} is chosen large enough to overcome the bounds of disturbances and parameter uncertainties.

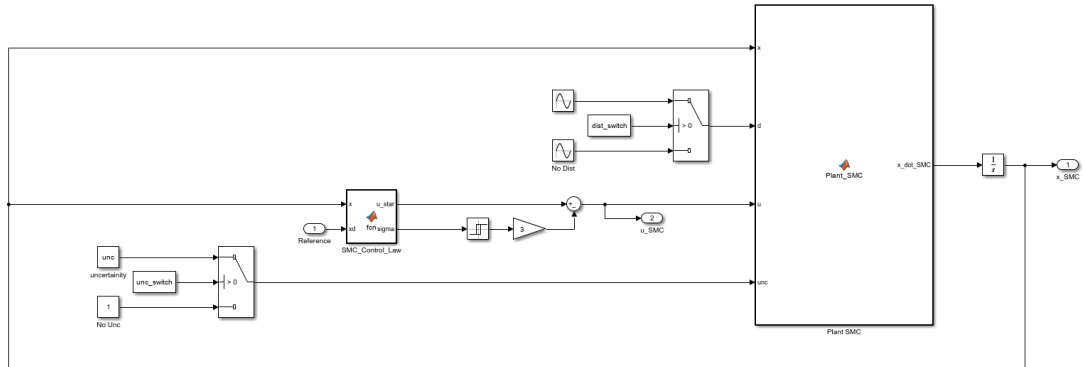


Figure 3.3: Simulink Block Diagram: Sliding Mode Controller

In detail the MatLab function showed in the SFC diagram to implement the control law, while the plant is the same used above for the SFC Controller.

```

1 function [u_star, sigma] = fcn(x, xd, unc)
2 % Nominal Parameters
3 J = 2.5e-4;
4 k1 = 0.15;
5 k2 = 0.5;
6 b1 = 0.01;
7 b2 = 0.001;
8
9 x1 = x(1);
10 x2 = x(2);
11
12 % Sliding Surface Parameters
13 % p1 determines the convergence rate of the error
14 p1 = 20;
15 p2 = 1;
16
17 % Sliding Surface Calculation
18 % sigma = p1 * error + p2 * error_dot (assuming xd_dot = 0)
19 sigma = p1*(x1 - xd) + p2*x2;
20
21 % Estimation of Resistive Torques
22 T_spring = k1 * x1 + k2 * x1^3; % Linear + Cubic
23 T_friction = b1 * x2 + b2 * x2 * abs(x2); % Viscous + Aero
24
25 % u_star computation:
26
27 % sigma_dot = p1*x2 + p2*x2_dot = 0
28 % p1*x2 + (p2/J) * (u_star - T_spring - T_friction) = 0
29
30 u_star = T_spring + T_friction - (J * p1 / p2) * x2;
31 end

```

Listing 3.2: MATLAB function for the SMC Control Law (Updated for the non-linear model)

3.3.2 Nominal Results

The performance of the SMC in the nominal case is reported below.

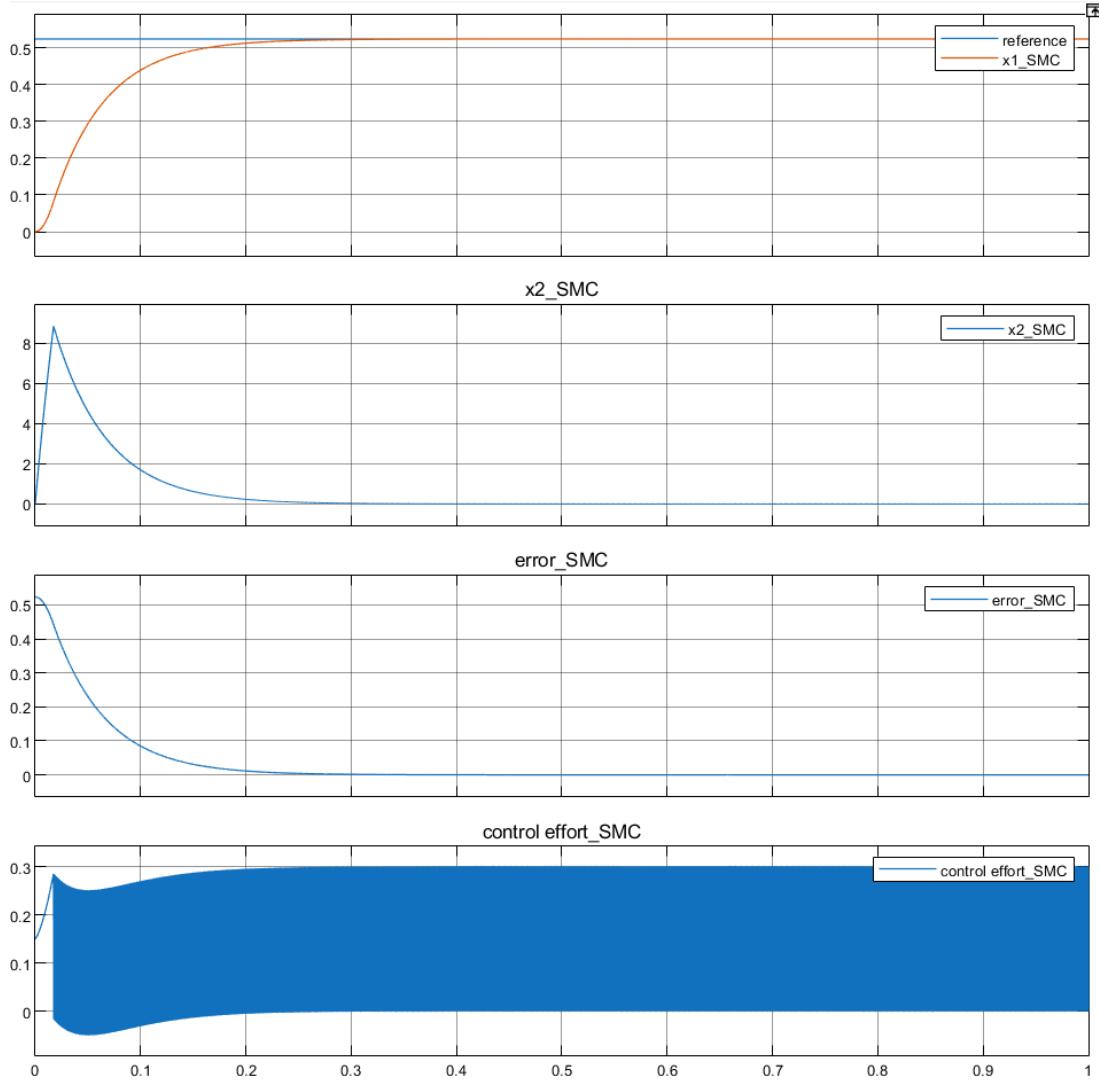


Figure 3.4: SMC Performance (Nominal): Top to bottom - Position tracking (x_1), Velocity (x_2), Tracking Error, Control Effort (u).

As seen in Figure 3.4, the SMC provides excellent tracking performance with the error converging rapidly to zero. The control effort plot highlights the characteristic **chattering** phenomenon (high-frequency switching) due to the discontinuous sign function, which is intrinsic to ideal Sliding Mode Control.

Chapter 4

Robustness Analysis

In this chapter, a robustness analysis of the previously synthesized controllers is carried out. The performance is evaluated first by introducing external disturbances, and subsequently by applying parametric uncertainties to the plant model. Finally, a worst-case scenario combining both disturbances and uncertainties is examined. The analysis is supported by simulations showing the evolution of the valve angle (x_1), velocity (x_2), tracking error, and control effort.

4.1 Robustness to Disturbances

The controllers are tested in a scenario where an external disturbance is added to the system input. Physically, this could represent airflow turbulence acting on the throttle plate or voltage fluctuations. In the simulation, a sinusoidal disturbance signal is added to the control input channel (matched disturbance).

4.1.1 State Feedback Controller (SFC)

The response of the State Feedback Controller is shown in Figure 4.1. Since the SFC is a linear controller designed without an integral action or a disturbance observer, the external sinusoidal disturbance explicitly propagates through the system dynamics. As visible in the plots, the output position x_1 oscillates around the reference value, preventing the error from converging to zero. The controller attempts to compensate, but the sinusoidal component remains present in the steady-state response.

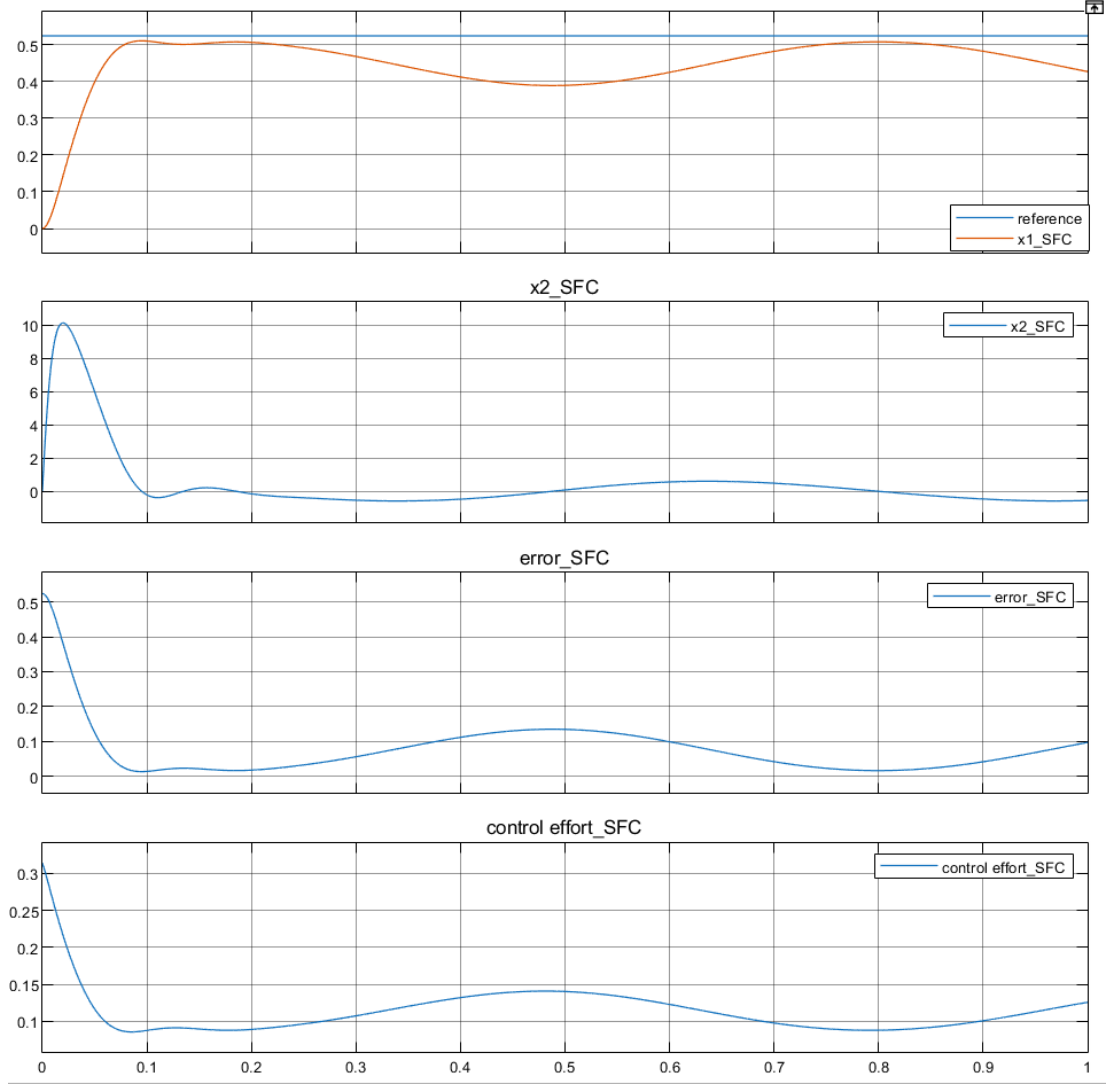


Figure 4.1: SFC under External Disturbance: Note the oscillations in the position tracking (x_1) and the non-zero error.

4.1.2 Sliding Mode Controller (SMC)

One of the fundamental strengths of Sliding Mode Control is its ability to reject disturbances, particularly when they satisfy the matching condition (i.e., they act in the same channel as the control input). The key mechanism is the discontinuous switching term u_{sw} . Once the system trajectory reaches the sliding manifold $\sigma(x) = 0$, the controller reacts immediately to any deviation caused by the disturbance. As shown in Figure 4.2, the system maintains excellent tracking performance. The state x_1 follows the reference with near-zero error, effectively canceling the effect of the sinusoidal disturbance.

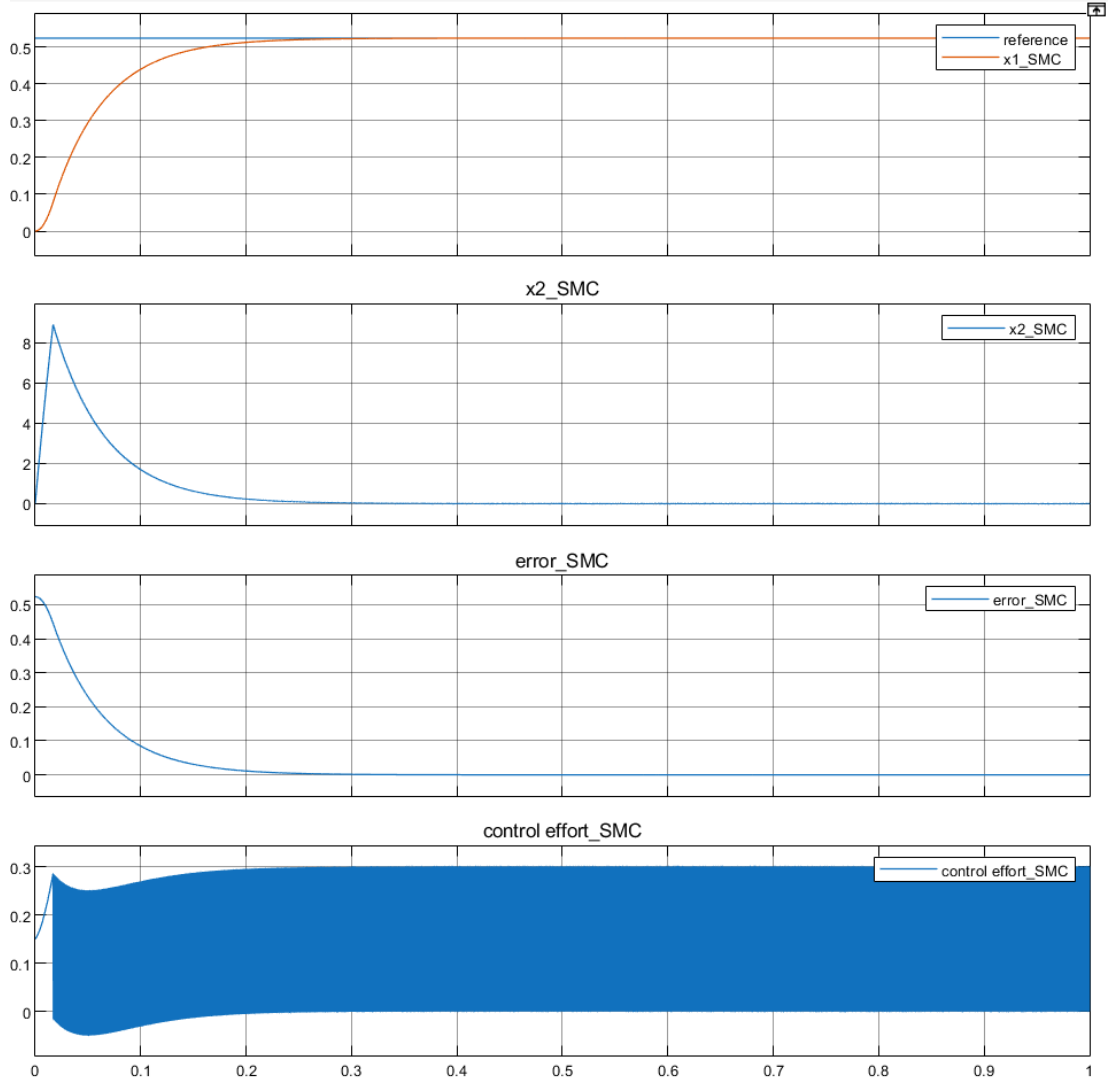


Figure 4.2: SMC under External Disturbance: The disturbance is completely rejected, resulting in perfect tracking.

4.2 Robustness to Uncertainties

The second case under analysis is the study of controller behavior in the presence of parametric uncertainties. This simulates modeling errors or physical changes in the component (e.g., spring wear, friction changes). For this test, the plant parameters in the simulation were altered relative to the nominal model used for control design:

- Inertia (J), friction (b_1) and stiffness (k_1, k_2) coefficients were increased by 30%.

4.2.1 State Feedback Controller (SFC)

The linear State Feedback controller relies on the precise cancellation of system dynamics via the feedforward gain K_r and the placement of poles based on the nominal matrices A and B . Figure 4.3 shows that under parametric mismatch, the SFC suffers from a noticeable degradation in accuracy. Specifically, a static steady-state error appears because the pre-computed feedforward gain is no longer valid for the altered physical plant.

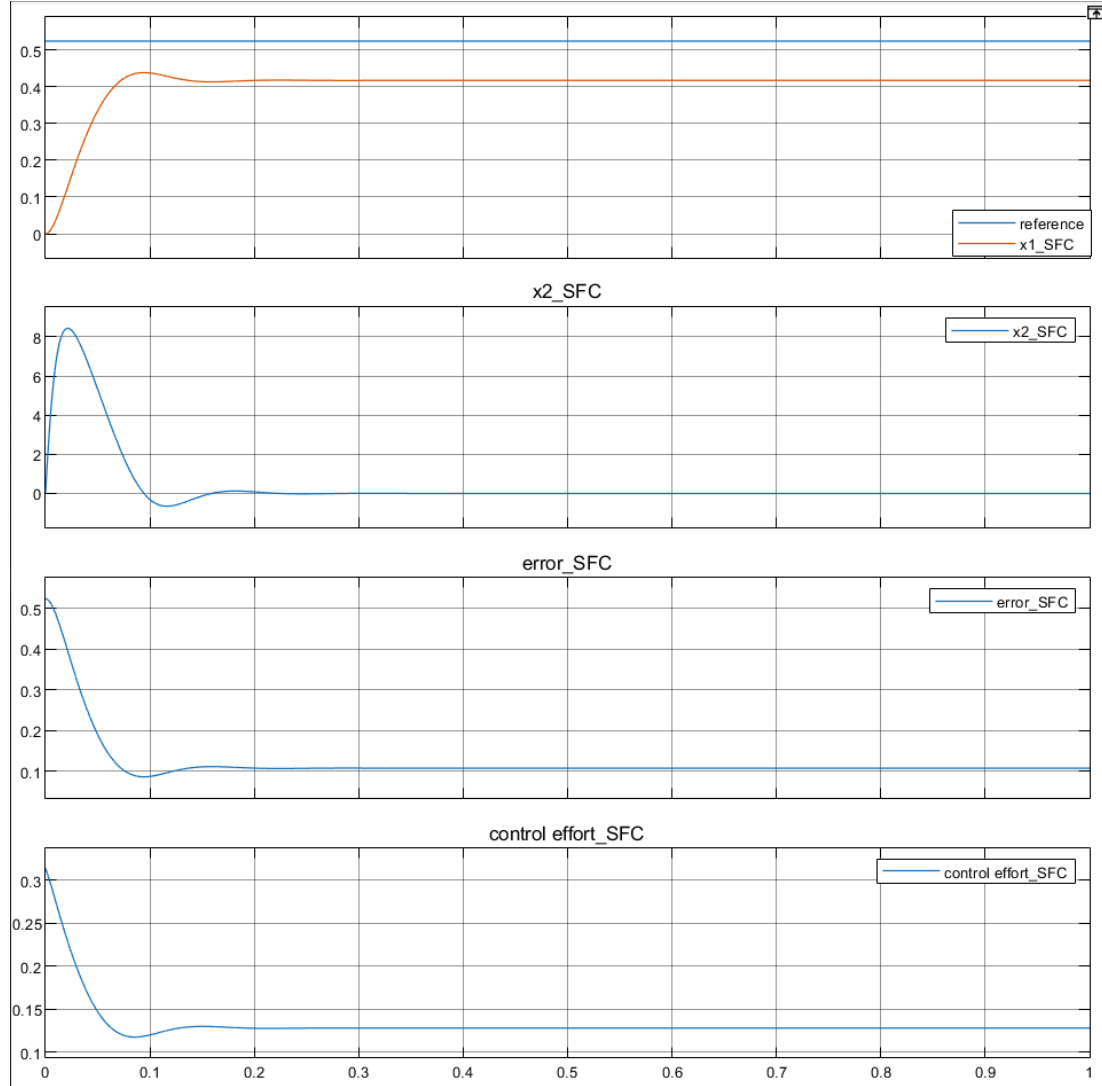


Figure 4.3: SFC under Parametric Uncertainty: A steady-state error (offset) is visible in the position graph.

4.2.2 Sliding Mode Controller (SMC)

The Sliding Mode Controller demonstrates superior robustness. As long as the switching gain K_{sw} is large enough to bound the uncertainties, the sliding condition $\sigma\dot{\sigma} < 0$ holds. Figure 4.4 confirms that the SMC is practically unaffected by the parameter variations. The output tracks the reference perfectly without any static error or significant transient degradation.

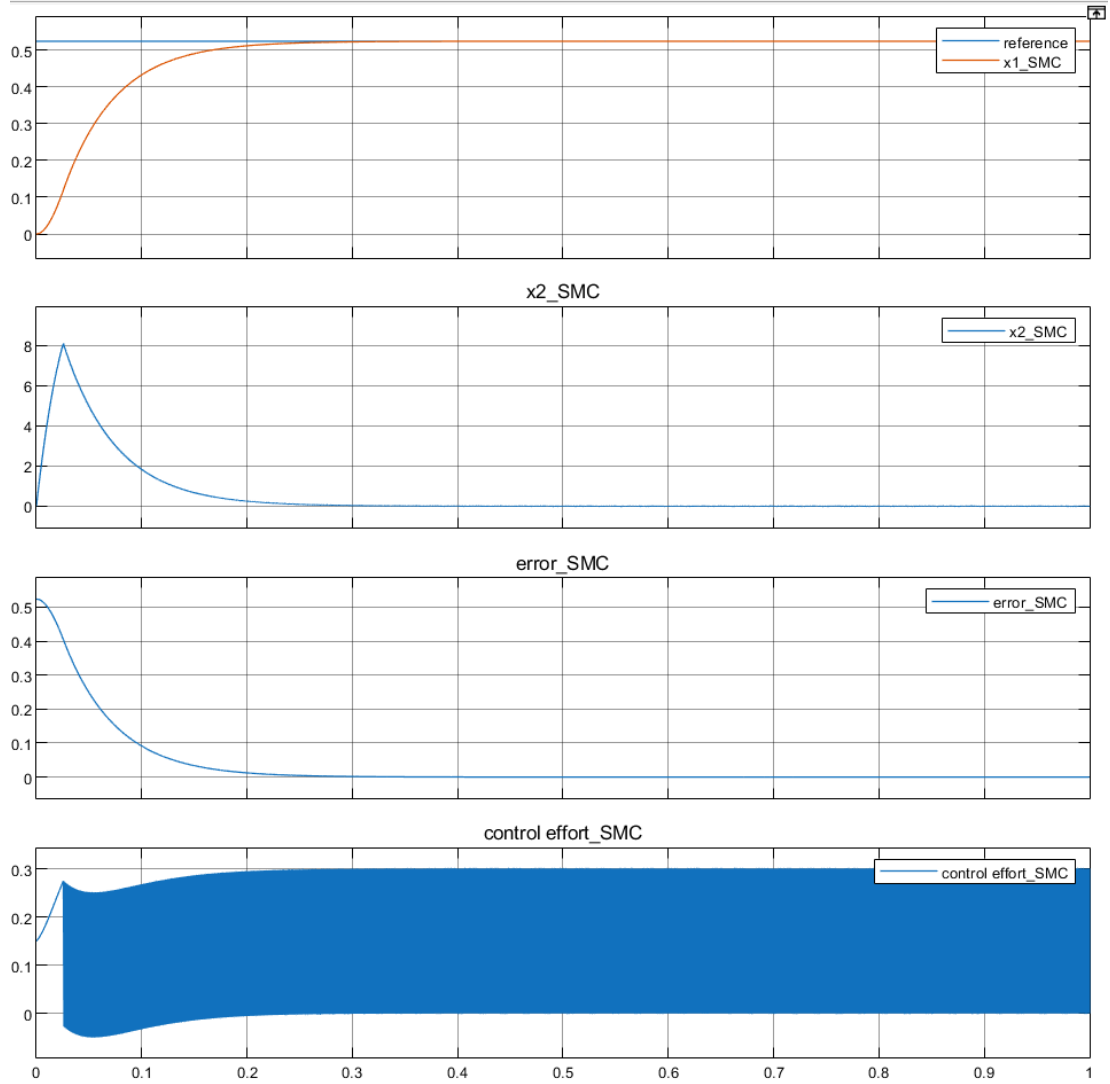


Figure 4.4: SMC under Parametric Uncertainty: The controller maintains robustness with no visible error.

4.3 Combined Scenario (Disturbance + Uncertainty)

Finally, a worst-case scenario is simulated where both the sinusoidal disturbance and the parametric uncertainties are applied simultaneously.

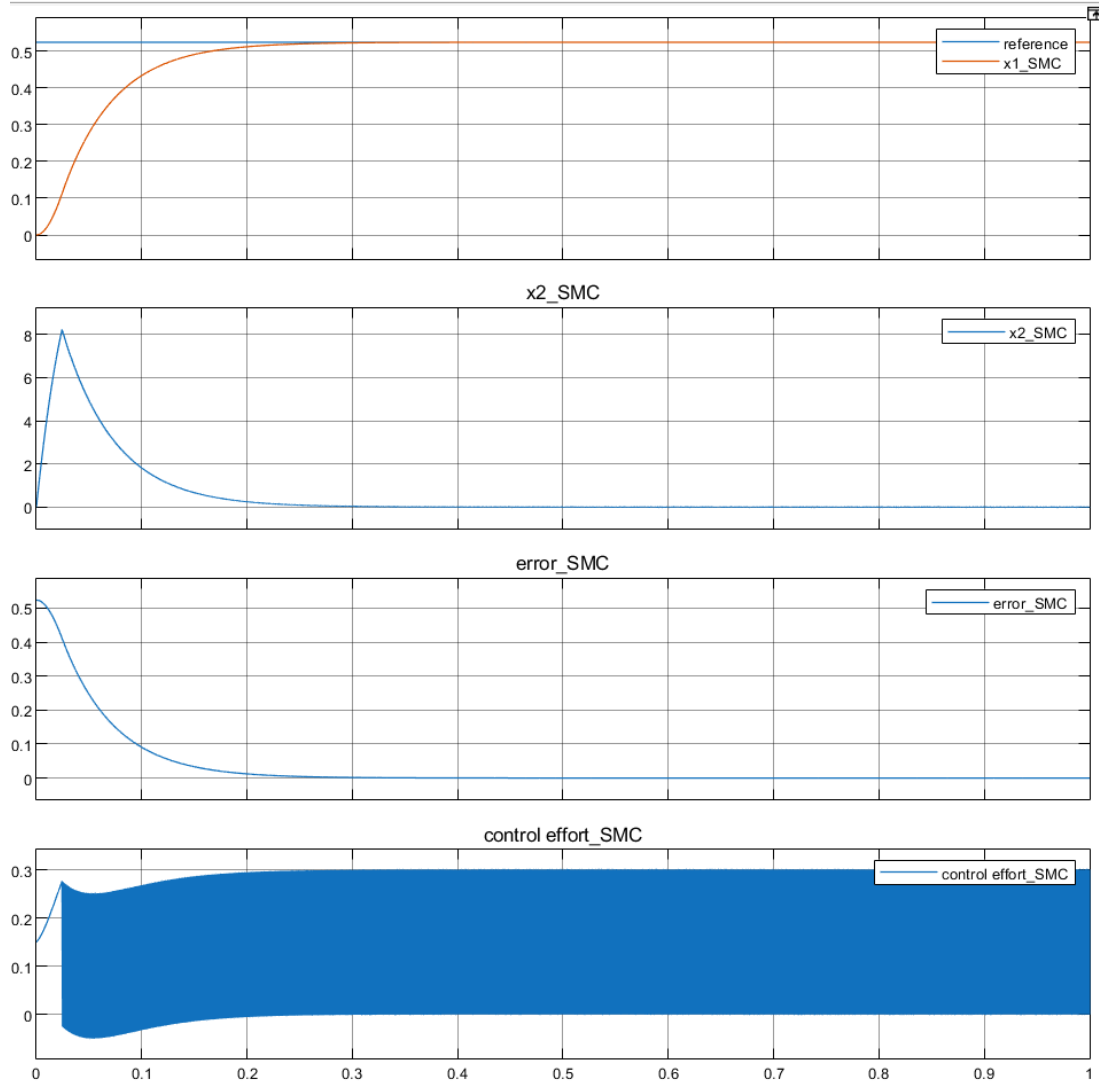


Figure 4.5: SMC under Combined Disturbance and Uncertainty: Robustness is preserved.

The comparison is striking. While the linear controller (not shown for brevity) would exhibit both oscillations and offset, the **SMC** (Figure 4.5) preserves its performance, proving to be a highly robust strategy for the Electronic Throttle Body application.

Chapter 5

Performance Comparison

Before concluding the report, a quantitative and qualitative comparison between the performance of the linear (SFC) and nonlinear (SMC) controllers is performed. This comparison highlights the trade-offs between implementation simplicity, tracking accuracy, and control effort.

5.1 Simulation Environment Setup

To ensure a fair and direct comparison, a unified Simulink model was developed. As shown in Figure 5.1, the architecture allows both controllers to run in parallel on identical plant models. A key feature of this implementation is the use of **Masked Subsystems**. Through editable masks, it is possible to intuitively inject external disturbances (defining amplitude and frequency) and apply parametric uncertainties (percentage variation of parameters) without modifying the underlying block diagram. This setup ensures that both controllers are subjected to the exact same rigorous testing conditions simultaneously. In addition, the external reference is editable using a manual switch in order to allow to insert a constant step or a signal built by sketch to test the controllers performances in dynamic tracking.

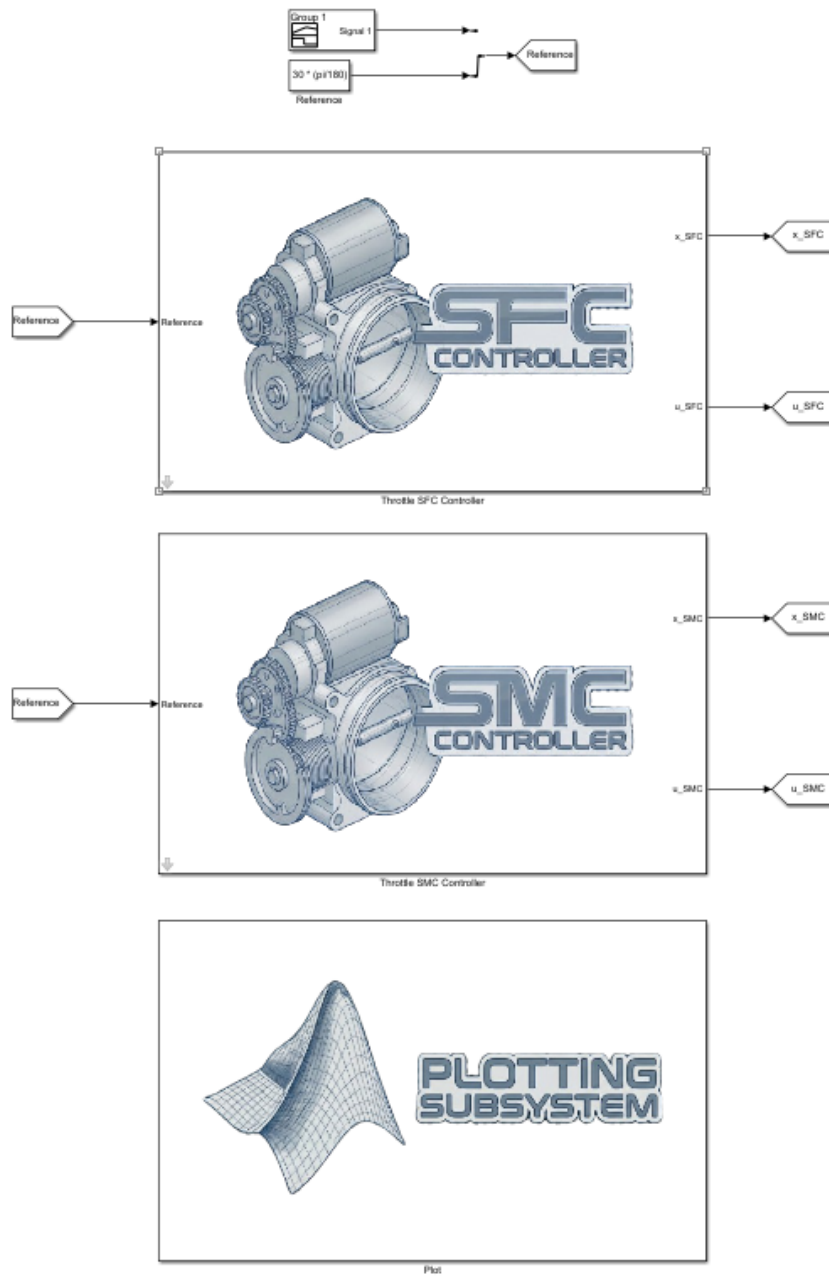


Figure 5.1: General Simulink Model Architecture: Parallel simulation of SFC and SMC with editable masks for Disturbance and Uncertainty injection.

5.2 Tracking Performance

5.2.1 Nominal Conditions

In the nominal case (Figure 5.2), both controllers manage to stabilize the system. However, a detailed inspection reveals the limitations of the linear approach. The **SFC** (red line), while smooth, exhibits a small steady-state error or offset relative to the reference. This is because the linear control law cannot perfectly compensate for the non-linear stiffness ($k_2\theta^3$) and aerodynamic damping ($b_2\dot{\theta}|\dot{\theta}|$) inherent in the throttle body. Conversely, the **SMC** (yellow line) forces the error to zero, achieving perfect tracking once the sliding surface is reached.

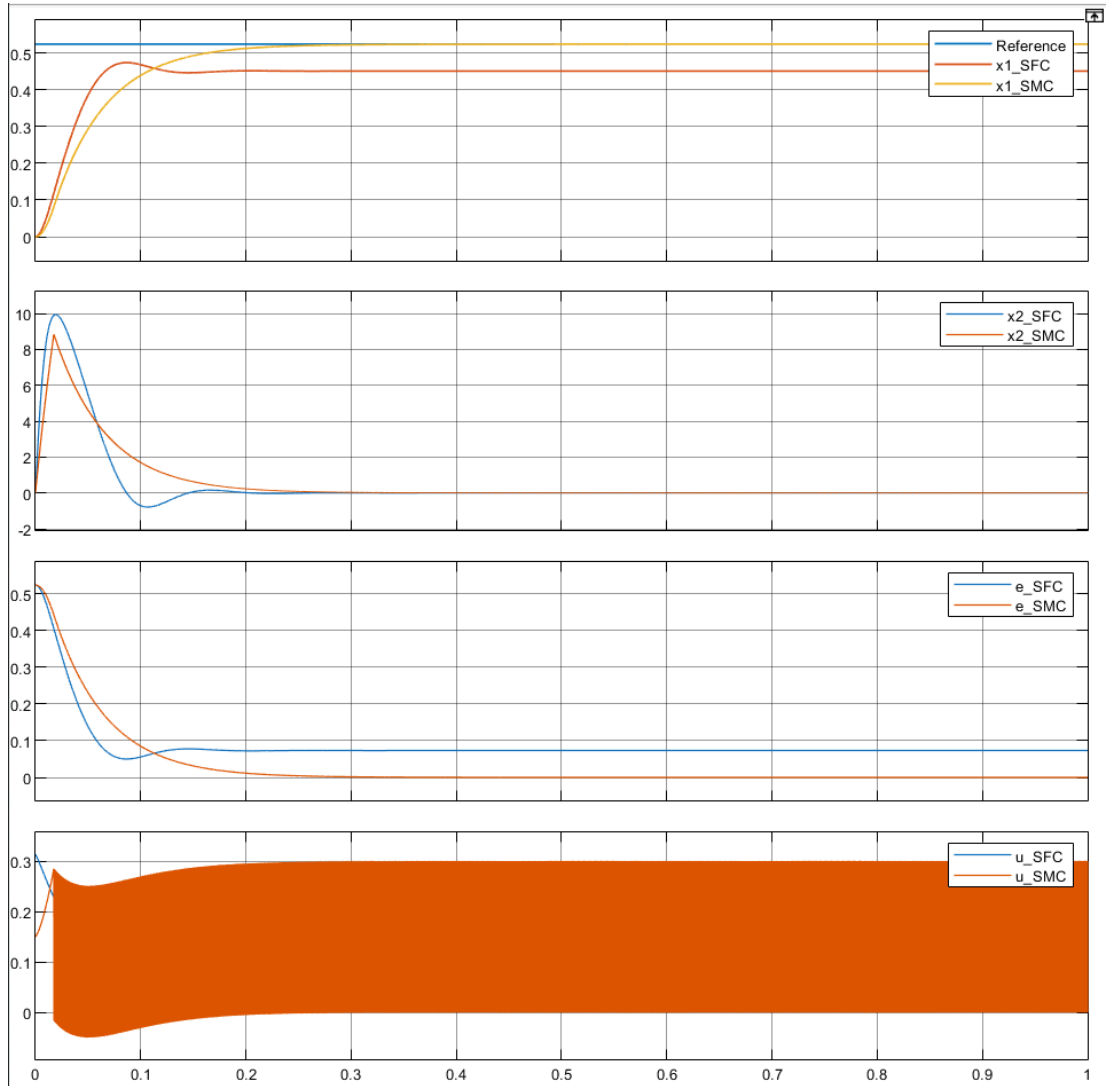


Figure 5.2: Comparison - Nominal Step Response: The SMC achieves zero error, while SFC shows a slight offset due to non-linearities.

Table 5.1: Quantitative Performance Comparison (Nominal Conditions)

Performance	SFC	SMC
Rising Time (T_r) [s]	≈ 0.03	≈ 0.11
Settling Time (T_s) [s]	≈ 0.11	≈ 0.2
Percent Overshoot ($OS\%$)	4%	0%
Steady-State Error (e_{ss}) [%]	$\approx 15.0\%$	0%

5.2.2 Robustness (Dynamic Tracking with Disturbances and Uncertainty)

The superiority of the non-linear approach becomes evident in the "worst-case" scenario. Figure 5.3 illustrates a dynamic tracking task (drive cycle similar reference) under the combined effect of sinusoidal disturbances and significant parametric uncertainty of 30%.

- SFC Performance:** The linear controller fails to track the reference accurately. The parametric mismatch causes a large static offset (the output is significantly lower than the reference), and the sinusoidal disturbance propagates directly to the output, causing visible oscillations. This is emphasized by the reference chosen that is far from the limp home position where the linearization is applied.
- SMC Performance:** The Sliding Mode Controller remains invariant to these perturbations. It tracks the dynamic steps with high precision, rejecting the disturbance and compensating for the model mismatch. The tracking error (e_{SMC}) remains practically zero throughout the simulation.

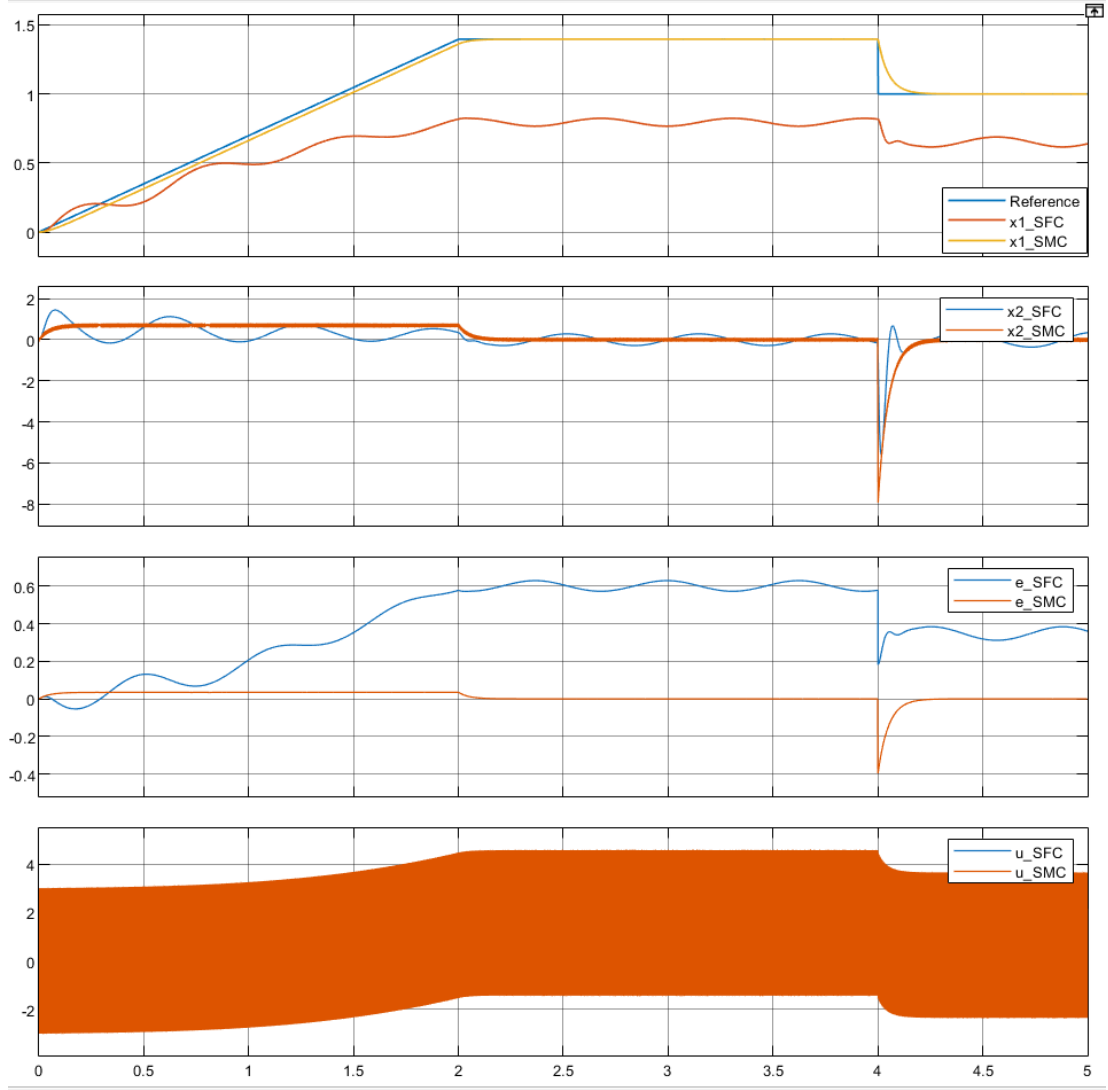


Figure 5.3: Comparison - Dynamic Tracking with Disturbance and Uncertainty: SMC demonstrates superior robustness compared to SFC.

5.3 All Scenarios

Although the tracking graphs compared in the nominal case with static reference and in the disturbance and uncertainty case with dynamic reference are already sufficient in themselves to show the necessary differences between the controllers, for the sake of completeness, the comparisons for all the other tested scenarios are reported below.

5.3.1 Disturbance with Static Reference

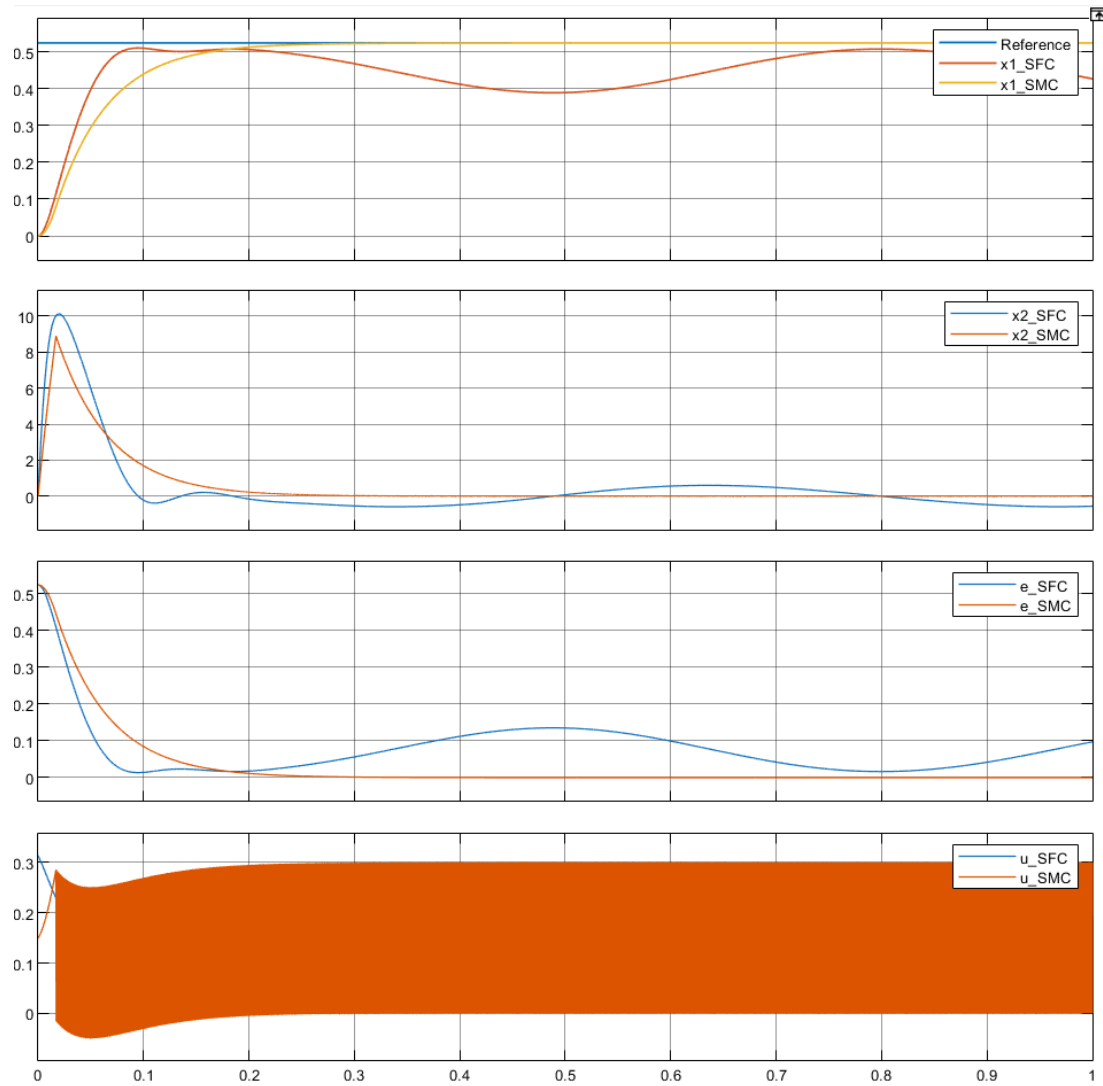


Figure 5.4: Comparison - Disturbance with Static Reference.

5.3.2 Uncertainty with Static Reference

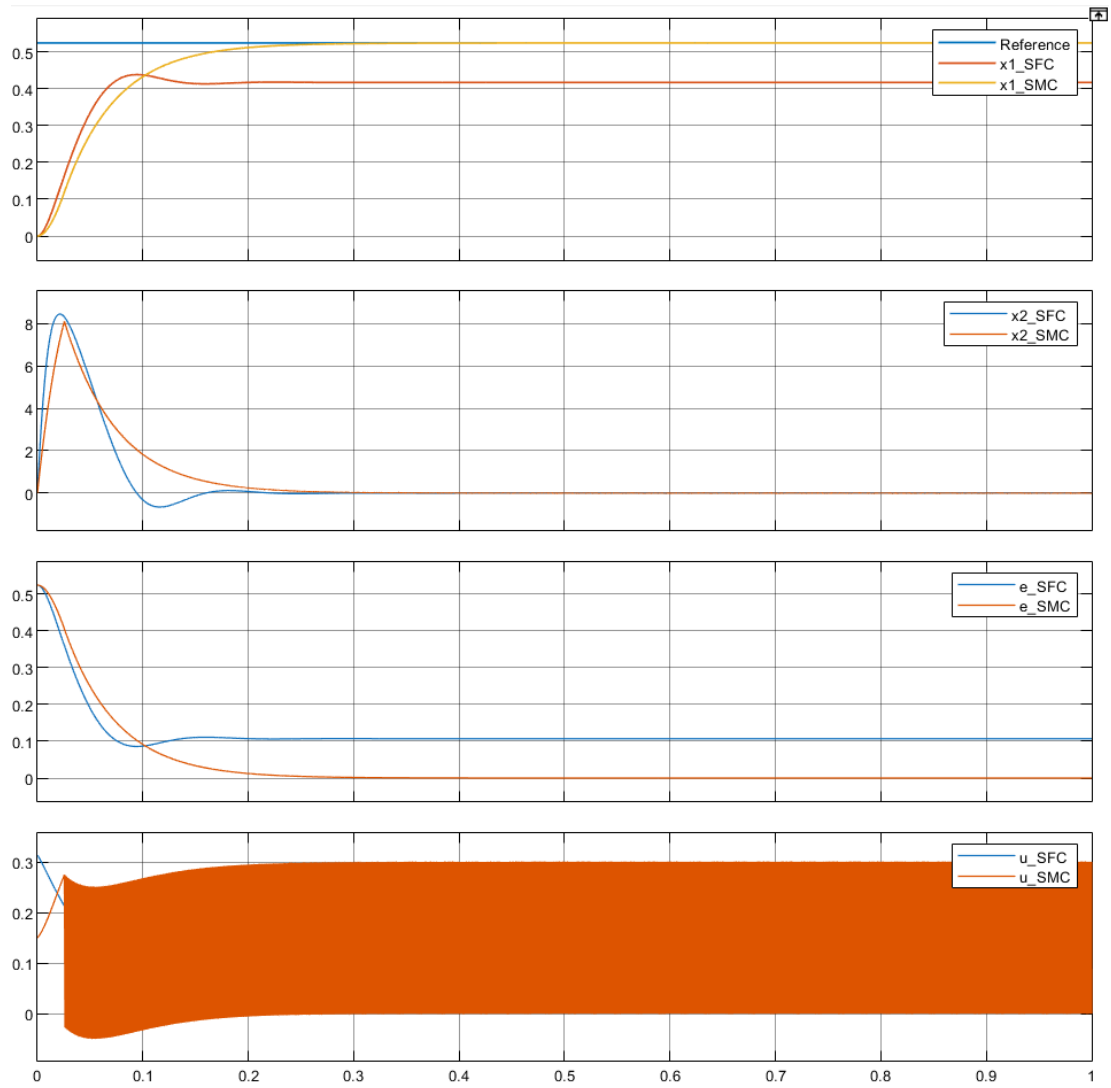


Figure 5.5: Comparison - Uncertainty with Static Reference.

5.3.3 Disturbance & Uncertainty with Static Reference

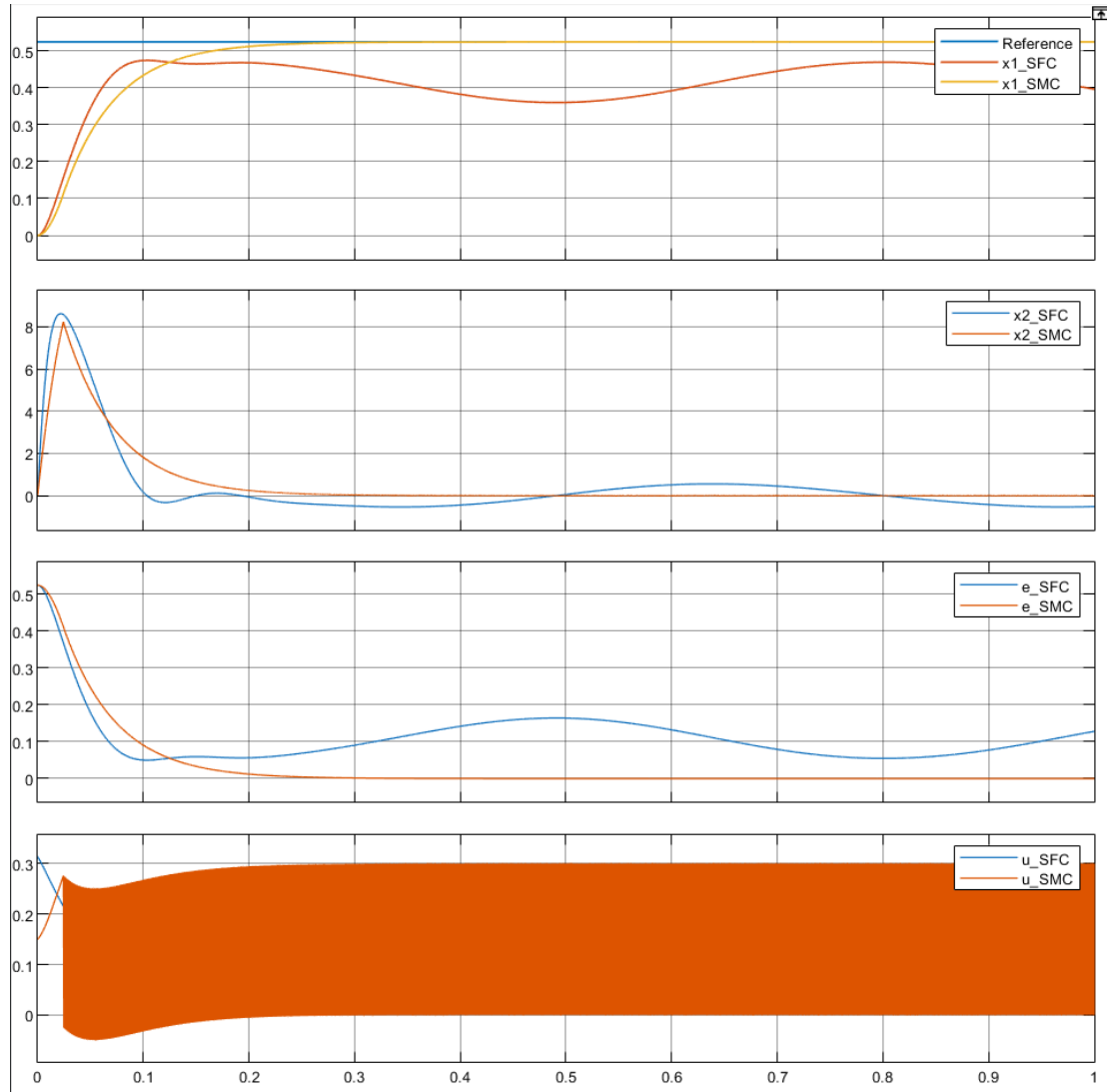


Figure 5.6: Comparison - Disturbance & Uncertainty with Static Reference.

5.3.4 Nominal with Dynamic Reference

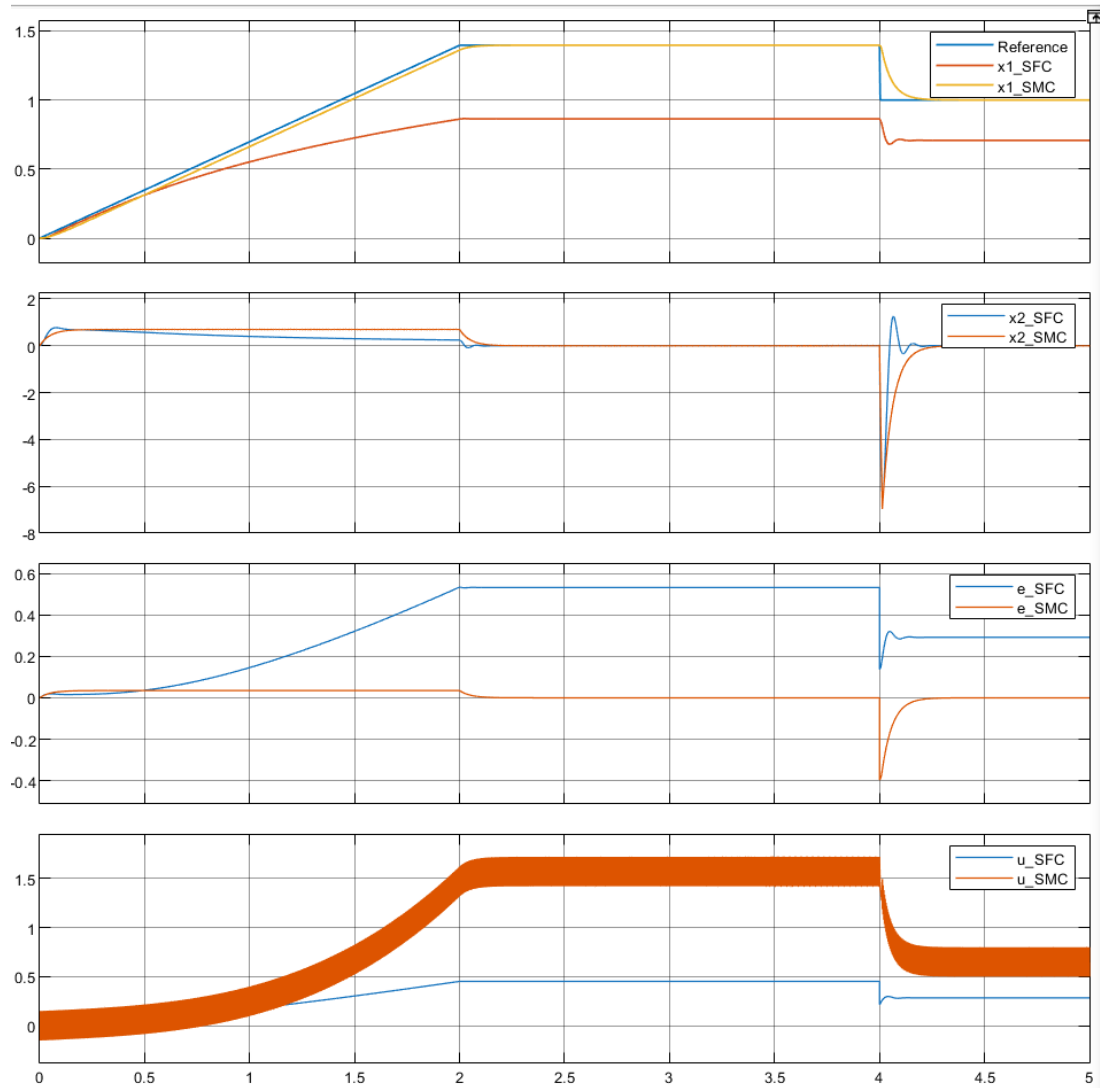


Figure 5.7: Comparison - Nominal with Dynamic Reference.

Chapter 6

Conclusion

In this report, the problem of modelling, analyzing, and controlling a non-linear dynamic system was addressed, with a specific focus on an **Electronic Throttle Body**. The study aimed to evaluate how non-linear phenomena influence system behavior and to compare the effectiveness of linear versus non-linear control strategies.

6.1 Summary of Analysis

The preliminary analysis, supported by the **Phase Plane** method, highlighted the strictly non-linear nature of the system. The presence of a cubic stiffness term and non-linear aerodynamic damping resulted in a system with a single stable equilibrium point (the origin). The analysis confirmed that while the system does not undergo stability bifurcations (Hopf) within the physical parameter range, it exhibits topological changes in its trajectories when damping or stiffness coefficients vary.

6.2 Control Strategy Comparison

Two control architectures were synthesized and tested in a unified Simulink environment using editable masks to inject disturbances and uncertainties:

1. **State Feedback Controller (SFC):** Designed on the linearized "Small Signal Model". It proved effective in nominal conditions, ensuring stability and reasonable transient response. However, its performance deteriorated significantly in the robustness analysis. The inability to compensate for non-linear dynamics led to steady-state errors under parametric uncertainty and visible oscillations in the presence of external sinusoidal disturbances.

2. **Sliding Mode Controller (SMC):** Designed explicitly to handle the system's non-linearities. The simulations demonstrated the superiority of this approach. In the "worst-case" scenario (combined disturbances and parametric mismatch), the SMC maintained perfect tracking capability ($e \approx 0$), proving its invariance properties.

6.3 Control Effort and Chattering

The performance advantage of the SMC comes at a cost, which is visible in the control input plots (u_{SMC} vs u_{SFC}).

As observed in the bottom subplots of the comparison figures, the SFC generates a smooth, continuous voltage signal, which is mechanically gentle on the DC motor and gears. On the other hand, the SMC relies on high-frequency switching to maintain the sliding condition. This results in the **chattering phenomenon**. While mathematically effective for robustness, in a real physical Throttle Body, excessive chattering can lead to:

1. Overheating of the electric motor coils;
2. Mechanical wear of the gears and the return spring;
3. Excitation of unmodeled high-frequency dynamics.

6.4 Conclusion on Comparison

In conclusion, the simulations confirm the theoretical superiority of the Sliding Mode Controller in terms of tracking accuracy and robustness against model uncertainties and external disturbances. The SFC, being a linear technique, is limited to a small operating range around the linearization point and lacks the authority to reject matched disturbances. However, for practical implementation, the SMC would require chattering alleviation techniques (such as replacing the *sign* function with a saturation or sigmoid function) to protect the actuators, creating a trade-off between absolute precision and mechanical durability.

Bibliography

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- [2] C. Rossi, A. Tilli, A. Tonielli. *Robust control of a throttle body for drive by wire operation of automotive engines*. IEEE International Conference on Control Applications, pp. 793-798, 2000.

Declaration of AI Usage

As far as this report is concerned, L^AT_EX was used for typesetting. In compliance with the guidelines, it is declared that Artificial Intelligence tools were utilized during the preparation of this document.

Specifically, the following tools were employed:

- **ChatGPT**: used for L^AT_EX command support, including formula syntax, document formatting, and arranging images (e.g., side-by-side figures).
- **Google Gemini**: used for the generation and processing of the mask images found in the implementation section.